

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EE) Examination
Dec 2012

EE 704 Advanced Power System Protection

Date: 21.12.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Discuss various components of digital relays used in power systems. 07

OR

Q – 1 Discuss various merits and demerits of digital relays, with reference to electromechanical and static relays. 07

Q – 2 A. Discuss Full cycle discrete Fourier algorithm with its advantageous. 07

B. Define wide area protection system. Explain three major design architectures of wide area protection system. 07

OR

A. Discuss half cycle discrete Fourier algorithm with its demerits. 07

B. Draw the basic block diagram of the synchronized phasor measurement unit and also explain the concept of synchronized sampling. 07

Q – 3 **Attempt any two** 14

(a) Define adaptive relaying and discuss its advantageous.

(b) Explain how the adaptive protection philosophy is achieved in power transformer.

(c) Explain various protection schemes used for busbar. How adaptive philosophy is helpful to rectify the problems in the conventional busbar protection scheme.

SECTION – II

Q – 4 Derive the impedance seen by the relay in the event of power swing. Explain the tripping characteristic of out of step tripping relay and set the criteria for its location 07

OR

Q – 4 Which are the bypass arrangements for capacitor bank? Discuss the relative advantages and disadvantages over each other. 07

Q – 5 (a) Find out the back up relays of Relay 6 for the system shown in Figure 1.

07

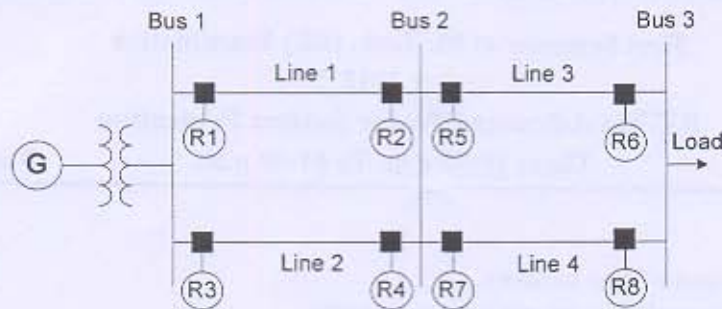


Figure 1

(b) Explain different criteria and procedure considered while designing load shedding scheme

07

OR

Q – 5 (a) List out the relaying problems occurred due to series compensation in transmission line and explain voltage and current inversion

07

(b) Explain the operating characteristic of different types of under frequency relays in details.

07

Q – 6 **Attempt any two**

14

- (1) List out the specification for current transformer and explain each. Also show the importance of burden selection on the accuracy of protective CT.
- (2) Explain the protection for voltage instability
- (3) Explain the criteria for deciding the location of the frequency relays for implementing load shedding scheme
